

## RESEARCH ARTICLE

# Performance Evaluation of a Novel Sequence-Based Directional Detection Strategy for Protection of Active Distribution Networks

KWASI OPOKU<sup>1</sup>, (Graduate Student Member, IEEE),  
ALEKSANDAR DIMITROVSKI<sup>1</sup>, (Senior Member, IEEE),  
AND MAXIMILIANO FERRARI<sup>2</sup>, (Senior Member, IEEE)

<sup>1</sup>Department of Electrical and Computer Engineering, University of Central Florida, Orlando, FL 32816, USA

<sup>2</sup>Grid Components and Controls Group, Oak Ridge National Laboratory, Oak Ridge, TN 37830, USA

Corresponding author: Kwasi Opoku (kwasi.opoku@ucf.edu)

This work was supported by the U.S. Department of Energy's Office of Energy Efficiency and Renewable Energy (EERE) through the Solar Energy Technologies Office under Award DE-EE0002243-2144.

**ABSTRACT** Directional elements are relied on to achieve selectivity in fault detection in power systems. Although such elements have been deployed successfully for many years, there is an increased need for novel methods to deal with the unique challenges of directional protection in modern distribution networks. This article analyzes the impact of inverter-based resources (IBRs) on existing directional protection methods in distribution systems. It identifies parts of such elements that pose a risk of misoperation when IBRs are used in distribution networks. The authors have developed a new directional detection method for unbalanced faults in such networks using superimposed symmetrical sequence quantities. The phase angle of the superimposed negative sequence admittance is used to determine fault direction. The paper also presents a real-time co-simulation platform between a simulated distribution system and physical protection relay, using OPAL-RT. An SEL-411L relay is used to program the detection algorithm. This hardware-in-the-loop (HIL) setup is used to verify the performance of the method and the results are compared with existing directional methods

**INDEX TERMS** Superimposed quantities, negative sequence, directional element, microgrid protection, HIL.

## I. INTRODUCTION

### A. BACKGROUND AND MOTIVATION

A major aspect of ongoing grid modernization efforts is the integration of distributed energy resources (DER). This allows locally-available resources, such as renewable energy sources, to be used to supply local loads. In this regard, such systems are also referred to as microgrids or active distribution systems (ADN) [1]. The integration of DERs has benefits in ensuring the resilience of the distribution system. However, ensuring reliable protection during faults in ADNs remains a major challenge and area of focus for many engineers and researchers. The

architecture of microgrids and the unique characteristics of the DERs, particularly inverter-based resources (IBR) such as solar and wind sources, depart from conventional distribution systems and introduce risk of protection failure in microgrids. The challenges to microgrid protection have been extensively discussed in existing literature [2], [3], [4], [5], [6], [7] and can be summarized broadly as follows

- 1) Fault current levels and characteristics for IBRs diverge from that of conventional sources. This unconventional fault behaviour impacts the sensitivity and speed of fault detection.
- 2) ADNs include distributed generation, varying modes of operation and complex topologies that impact the radial architecture of conventional distribution systems. This

The associate editor coordinating the review of this manuscript and approving it for publication was Ahmed A. Zaki Diab<sup>1</sup>.

potentially affects the selectivity of legacy distribution protection.

An effective microgrid protection method requires accurate determination of fault direction. Directional protection is, however, significantly impacted in the operation of such networks due to the bidirectional flow of fault currents. Coupled with the unconventional fault signals, this impacts the accuracy of directional elements used in existing commercial relays. Also, the distributed nature of such sources, poses the risk of protection blinding and sympathetic tripping situations [8] during faults.

## B. STATE OF THE ART

In order to deal with the challenge of directionality, some proposed solutions depend on distance relay and differential relay principles used in transmission networks [9], [10], [11], [12]. Directly adopting transmission principles faces a number of drawbacks. For instance, distance protection in a distribution system must consider infeed from different DERs and the risk of overreach due to proliferation of lateral lines (These may require a higher number of relays to resolve). Also, communication infrastructure for pilot schemes is needed. Differential protection in particular, while effective, requires high-speed communication and additional backup protection in the event of communication loss. These increased investments make such methods unattractive for distribution systems.

Directional overcurrent relay (DOCR) principles used in conventional networks remains the most common approach to dealing with the challenge of bi-directional current flow in microgrids [13], [14]. The traditional approach relies on overcurrent detection and a phase shift between a chosen polarizing and operating quantity to determine direction. But, as previously mentioned, in order for such conventional DOCRs to be employed in microgrids with IBRs, the major differences in fault characteristics and topologies have to be accounted for. As shown in [12] and [15], such methods that rely only on fault currents are impacted by control schemes of IBRs as well as other factors in the microgrid, rendering the schemes unreliable. Furthermore, as identified in [16], [17], and [18], the conventional sequence-based directional methods are also affected since the underlying assumptions for phase relationship between the quantities may not hold for IBRs. A common solution to this is to employ adaptive DOCR relays that rely on communications to adjust protection operation based on changes in system conditions. As discussed in [13], the effectiveness of adaptive DOCRs depends on the ability to differentiate inverter fault currents from regular occurrences like peak load and motor starting currents.

Some proposed solutions attempt to address the low fault current condition faced by DOCRs by focusing on the coordination challenge arising from this condition. In this regard, various conventional and heuristic methods are used to optimize the time multiplier and pickup (plug multiplier)

settings for the overcurrent feature [19], [20], [21], [22]. These methods however do not address the underlying fundamental challenge of the direction of incident fault and how the low fault current impacts it.

An important area of focus in recent literature, therefore, is an attempt to develop new directional elements from fault quantities that are different from existing methods in commercially available relays. Such methods include the use of harmonic current quantities [23], [24], [25], a comparison of different faulted phase angle measurements [26], [27] as well as monitoring of transient signals [28], [29]. Similarly, many artificial intelligence based methods have been proposed for directional determination [30], [31]. The applicability of all these methods hinges on other factors such as communication infrastructure, ease of computation, and integration with the existing power network.

Similarly, new approaches also focus on novel inverter technologies to emulate conventional characteristic fault signatures. This is aimed at avoiding misoperation of both conventional and novel relay technologies. Such methods may emphasize new control strategies for inverter behaviour during faults. References [32], [33], and [34] include fault-ride-through (FRT) methods that employ dual current control loops for both positive and negative sequence currents ( $I_1$  and  $I_2$ ). In this case, the inverters emulate synchronous generator behaviour during balanced and unbalanced faults, respectively. The approach in [35] also introduces a new current control to determine appropriate current references to prevent misoperation of distance relays during unbalanced faults. Alternatively, in [36], hardware modifications and control strategies are proposed for grid-forming (GFM) inverters in order to ensure both a significant increase in short-circuit current and control stability during faults. This also improves the overcurrent requirement of conventional DOCRs. While this emphasis on obtaining appropriate fault quantities is vital for modernizing the grid, it may not be applicable for already installed and operational IBRs. In this context, seamless setting changes to existing network components offer desirable applicability.

## C. CONTRIBUTION

The contribution of this paper is three-fold. First, the paper discusses a new directional method for detecting asymmetrical faults in microgrids. The scheme employs superimposed negative sequence admittance measurements to detect the faults, and determine the fault direction. Furthermore, the paper demonstrates the verification of the scheme through real-time HIL simulation in a laboratory environment. The setup includes a modified version of the IEEE-13 node test feeder (modeled in MATLAB environment) and simulated in real-time using OPAL-RT. This way, the paper provides insights into the setbacks faced by commercially available directional protection algorithms. Finally, a practical approach to implementing the algorithm in the SEL-411L relay is presented and this is integrated into the testbed via

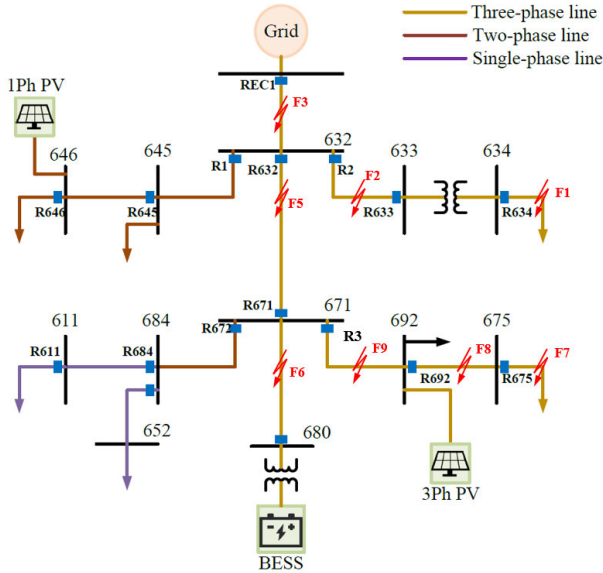


FIGURE 1. Modified IEEE 13 node test feeder.

DNP3 communication protocol. This provides a possible seamless approach to integrating the method into existing relays without the need for firmware upgrades. A comparison between the new method and existing directional methods in commercial relays is also presented.

The rest of the paper is organized as follows; Section II briefly describes the test system used for evaluation, and the control systems of the IBRs. Section III reviews some commonly employed directional elements and their unique challenges in the presence of DGs. In this regards, this section serves at the problem statement. Section IV presents the superimposed negative sequence admittance strategy as the proposed solution. Section V outlines the laboratory test setup used for implementing the proposed protection scheme and Section VI discusses the case study and results. Section VII concludes the paper.

## II. TEST SYSTEM

A modified version of the IEEE 13-node test feeder, shown in Fig. 1, is used to analyze the performance of different types of directional protection methods. The benchmark test feeder is modified to reflect the protection challenges introduced by the integration of IBRs. The grid side of the network is interfaced via 115kV/4.16kV substation with 5000kVA rating. The three IBRs included in this network are

- a 250kW Battery Energy Storage System (BESS) connected at bus 680
- a 125kW three-phase solar PV connected at bus 692
- a 40kW single-phase solar PV at bus 646

Two modes are considered in this microgrid. In the grid-connected mode, all the IBR inverters operate as grid-following (GFL) and the voltage and frequency references are established by the main grid. In the islanded mode, the BESS

inverter operates as grid-forming (GFM) and determines the voltage and frequency for the grid. The inverter controls developed in the microgrid controller in [32] and [33] are adopted for this system to emulate practical scenarios. Details are presented in Section II-A. The line characteristics and loads are kept the same as in the original feeder. Relays are positioned at various bus locations to maximize the reliability of the network. Fault locations shown in Fig. 1 are used for the analysis and performance verification in subsequent sections.

### A. CONTROL FUNCTIONS OF IBRs

#### 1) GRID-CONNECTED MODE

In the grid-connected mode, the IBRs, including the BESS operate as GFL to provide control of active power (P) and reactive power (Q) output. The current reference in  $dq$  axis,  $I_d^{ref}$  and  $I_q^{ref}$  for the predetermined P and Q reference points,  $P_{IBR}^{ref}$  and  $Q_{IBR}^{ref}$  are given by the formulation in (1).

$$\begin{aligned} I_d^{ref} &= \frac{V_q P_{IBR}^{ref} + V_d Q_{IBR}^{ref}}{V_d^2 + V_q^2} \\ I_q^{ref} &= \frac{V_d P_{IBR}^{ref} - V_q Q_{IBR}^{ref}}{V_d^2 + V_q^2} \end{aligned} \quad (1)$$

where  $V_d$  and  $V_q$  represent components of the grid voltage.

A fault-ride-through (FRT) is also implemented in the control system with a limited overcurrent capability. A drop in the grid voltage leads to increase in  $I_d^{ref}$  and  $I_q^{ref}$ , as can be seen in (1). This is used to model the FRT according to the rated maximum output of the IBR. Formulation (2) provides the new reference when the output current exceeds the IBR maximum during the fault.

$$\begin{aligned} I_{df}^{ref} &= \frac{I_{IBR}^{max}}{\sqrt{(I_d^{ref})^2 + (I_q^{ref})^2}} \cdot I_d^{ref} \\ I_{qf}^{ref} &= \frac{I_{IBR}^{max}}{\sqrt{(I_d^{ref})^2 + (I_q^{ref})^2}} \cdot I_q^{ref} \end{aligned} \quad (2)$$

where  $I_{IBR}^{max}$  represents the maximum output of the IBR and  $I_{df}^{ref}$  and  $I_{qf}^{ref}$  are the new current references during a fault.

#### 2) ISLANDED MODE

In this mode, the BESS provides the voltage and frequency for the network through a primary and secondary droop control. The frequency is determined by controlling the P in a P-f droop, while the output voltage is determined by controlling Q in a Q-V droop. A virtual impedance control is also implemented. Since the assumption in a droop control that the line reactance significantly exceeds line resistance may not hold true in a microgrid (MG) with a low X/R ratio, inaccuracies may arise in the P-f and Q-V droop relationships. To address this, virtual impedance control is applied to the BESS controller.

A dual loop control structure in  $dq$  coordinates is used for the voltage. The outer loop controls the inverter's output

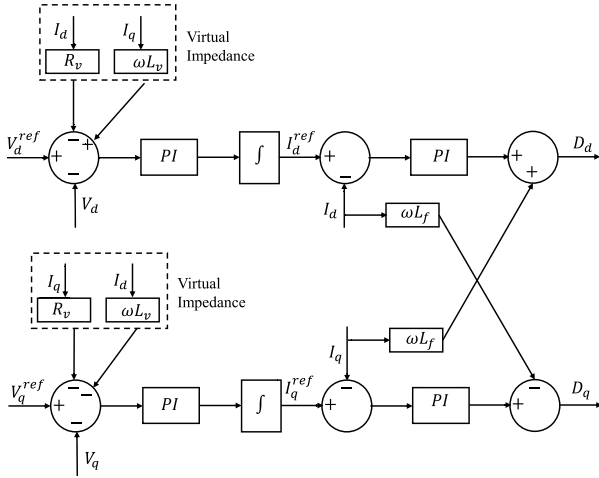


FIGURE 2. Islanded mode dual loop control structure for GFM.

voltage to set current references, while the inner loop adjusts the output current to match these references. This is illustrated in Fig. 2.

Under fault conditions, the FRT characteristic of the BESS controller restricts its output current to the predetermined allowable maximum. The fault current reference is generated according to (2). However, in this mode, the current references  $I_d^{ref}$  and  $I_q^{ref}$  are determined by the outer voltage loop, and not by (1). After the BESS fault-ride-through, it is unable to sustain the output voltage, resulting in a low-voltage condition for the system.

### B. SEQUENCE QUANTITY CONTRIBUTION

Similar to Section II-A, the current limiting and sequence contributions approach for IBRs modeled in this test case are based on developed controllers in [32] and [33]. In grid-connected mode, the negative sequence quantity contributions is determined by the main grid. The IBRs in this case operate in GFL mode, thus contributing only positive sequence currents to their rated maximum. In islanded mode, the GFM inverter regulates the positive and negative sequence components separately. The same dual-loop control structure in  $d - q$  coordinates (shown in Fig. 2) is applied for the control in both positive and negative sequence. In each case, the  $V_d^{ref}$  and  $V_q^{ref}$  are the respective positive and negative sequence voltages. In the event of a fault, reactive  $I_2$  is contributed based on the negative sequence voltage  $V_2$  developed to support the stability of the grid. This is similar to requirements provided in the German grid code VDE-AR-N 4120 [37]. Unlike [37], however, the magnitude of  $I_2$  contributed is not based on a pre-defined factor, but on a conservative current limiting algorithm for the overall inverter phase current output, explained in the next paragraph.

The current magnitude limiting strategy for negative and positive sequence currents is as shown in Fig 3, where  $I_{1max}$  and  $I_{2max}$  are the maximum allowed  $I_1$  and  $I_2$  respectively.  $I_{1f}$  and  $I_{2f}$  are the magnitudes of the faulted sequence currents

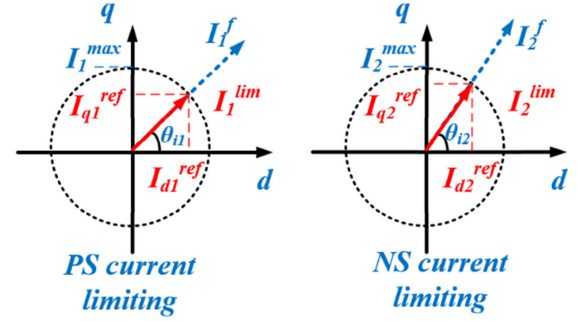


FIGURE 3. Current limiting strategy.

and  $I_{d,q1}$  and  $I_{d,q2}$  are the sequence currents in  $d - q$  coordinates. To limit the output phase currents of the three-wire inverters, both  $I_1$  and  $I_2$  must be limited. In this regard, a conservative limitation strategy is chosen to obtain the maximum inverter output current,  $I_{invmax}$ . This gives us:

$$I_{invmax} = I_{1max} + I_{2max} \quad (3)$$

A summary of the algorithm used to determine  $I_1$  and  $I_2$  levels in the FRT is provided below. The ride-through strategy attempts to maximize the healthy phase voltages without exceeding temporary overvoltage requirements. The sequence quantities are determined as follows:

- 1) To maintain current limiting, the maximum current should not exceed the sum of maximum negative and positive sequence current:  $I_{max} \leq I_{1max} + I_{2max}$ ;
- 2) During normal conditions,  $I_{1max}$  should be 1pu and  $I_{2max}$  should be equal to the  $I_2$  of the source;
- 3) If the sum of  $V_1$  and  $V_2$  is greater than 1pu, then  $I_1$  and  $I_2$  approach the same value;
- 4) If the sum of  $V_1$  and  $V_2$  is less than 0.8pu, then  $I_{1max}$  is increased while  $I_{2max}$  is reduced;
- 5) If  $I_{1max}$  is greater than 1.5pu or  $I_{2max}$  obtained is less than 0, the value is held to maintain the low short-circuit characteristics.

### III. OVERVIEW OF DIRECTIONAL PROTECTION METHODS

This section provides a brief overview of different directional protection methods and how these methods are impacted by an active distribution network. Many directional methods are based on the generic phase directional protection implemented in electromechanical relays using a generally constant reference quantity and the faulted current as operating quantity [38]. A general form of the torque,  $T$ , (based on A-phase polarizing current,  $I_a$ , and the BC line-line reference voltage,  $V_{bc}$ ) is obtained as

$$T = K \cdot |I_a| \cdot |V_{bc}| \cdot \cos[\theta - 90^\circ] \quad (4)$$

where

$K$  is a design constant based on the characteristics of the relay

$\theta$  is the phase angle between  $I_a$  and  $V_{bc}$



This element declares a forward fault for a positive torque, and reverse fault for a negative torque. Similar expressions are developed for B-phase and C-phase. This approach was however known to possess the risk of failure during ground faults with remote zero-sequence infeed [39]. Sequence-based methods were thus favored above these in later relay developments.

#### A. POSITIVE SEQUENCE DIRECTIONAL ELEMENT

The operation of the positive sequence directional element typically involves assessing the phase angle between positive sequence current and voltage. This is used to supervise the overcurrent element to form the DOCR. A common application employs a torque-based methodology [40], reflecting its initial adoption in the electromechanical relay and it can be expressed as

$$T32P = |V_1| \cdot |I_1| \cdot \cos[\angle V_1 - (\angle I_1 + \angle MTA)] \quad (5)$$

where

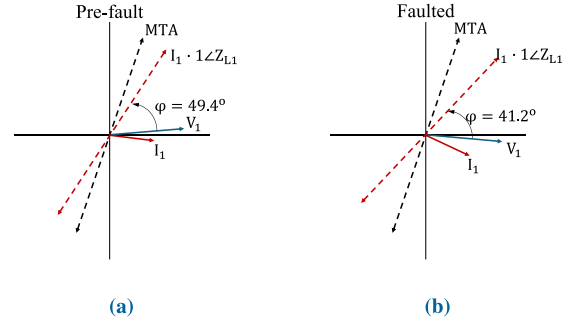
$T32P$  is the torque product of the directional element

$V_1$  and  $I_1$  are the positive sequence voltage and current respectively

$MTA$  is the characteristic angle of the line, typically the positive sequence impedance angle of the protected line ( $Z_1$ )

From (5), the correct operation of  $T32P$  will result in a positive sign for forward faults and negative for reverse faults. In other words, the angle between  $V_1$  and  $(I_1 \cdot 1\angle Z_1)$  should be  $-90^\circ$  to  $90^\circ$ . The torque calculated must exceed a minimum threshold for a valid directional decision to be made. A positive sequence memory voltage ( $V_{1mem}$ ) is frequently employed in modern microprocessor-based relays as the polarizing quantity to prevent element failure during close-in faults, where voltage magnitudes approach zero and the minimum positive sequence torque is not reached [41].

This element may however be affected by an unbalanced out-of-section fault, especially with fault currents approaching the load angle. For instance, consider a double-line-to-ground (2LG) bolted fault involving phases B and C, at F1 in Fig 1. In this scenario, the fault is reverse for R632 and the relay should not trip for the fault. However, the  $T32P$  element misoperates as it returns a torque product of +188 and identifies the fault in the forward direction. During normal grid-connected operation, the load flow is top to bottom (bus 650 to bus 671) with  $V_1$  of  $104.4\angle 5.6V_{sec}$  (on a VT ratio of 4000:120) and  $I_1$  of  $5.5\angle -16.4A_{sec}$  (on a CT ratio of 300:1). Hence, the normal operating positive sequence angle is heavily influenced by the grid supply. The  $MTA$  for this zone is  $71.4$ . During the fault, a minor portion of the total active power is supplied by the IBRs due to significantly reduced voltage and the lower  $I_1$  contribution to the faults by IBRs. The active power to loads is predominantly from the grid, thus the angle between  $V_1$  and  $(I_1 \cdot 1\angle Z_1)$  is  $41.2^\circ$ , within the positive cosine range as shown in Fig. 4.



**FIGURE 4.** Relay R632 positive sequence measurement during B-C-G fault at bus 634 (a)Pre-fault state (b)Faulted state.

#### B. NEGATIVE SEQUENCE DIRECTIONAL ELEMENT

Negative sequence directional elements have been widely adopted in microprocessor-based relays due to the ease of obtaining such quantities which are not much influenced by the load. They provide directional supervision for unbalanced faults. In this regard, the operation of this element typically overrides the decision of the positive sequenced based element during such unbalanced faults.

##### 1) TORQUE-BASED NEGATIVE SEQUENCE DIRECTIONAL ELEMENT

An early implementation of the negative sequence approach was a torque-based method, similar to  $T32P$ , that follows the operation of the electromechanical relay [42]. The torque equation is given as

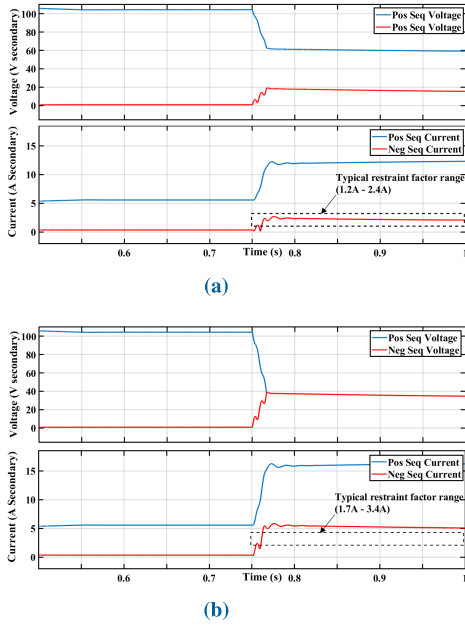
$$T32Q = |V_2| \cdot |I_2| \cdot \cos[\angle -V_2 - (\angle I_2 + \angle Z_{L2})] \quad (6)$$

$T32Q$  is the torque product of the directional element

$V_2$  and  $I_2$  are the negative sequence voltage and current respectively

$MTA$  is the characteristic angle of the line, as in (5)

For negative sequence quantities (unlike positive sequence quantities), a forward fault is declared when the angle between  $V_2$  and  $I_2$ , is  $180^\circ$  (minus the maximum torque angle). The negative sign in front of  $V_2$  is a reversal of the polarizing voltage angle ( $1\angle 180^\circ$  shift) in order to achieve a positive cosine for forward faults. Like other torque-products, therefore, it can be realized from (6) that for fault conditions with low  $V_2$  and/or  $I_2$  measurement, the  $T32Q$  could be too low to overcome the minimum torque setting. It is known that  $|V_2|$  from a remote fault has an inverse relationship with the strength of the source behind the relay [42]. Thus, under such source conditions, and with fault resistance that reduces  $|I_2|$ , the sensitivity of this element can be severely challenged. For example consider a B-C-G fault at F9 in the grid-connected mode. In this case, Relay R632 should see the fault in the forward direction and may serve as upstream backup to isolate line 632-671. However, the  $|V_2|$  recorded by R632 for this remote fault is a low voltage of  $11.5V_{sec}$  due to the stronger grid behind the relay. At severely low voltages, the angle of the voltage becomes unreliable for the element.



**FIGURE 5.** Measured  $V_1$ ,  $V_2$ ,  $I_1$ , and  $I_2$  by relay R632 for a) a remote fault at F9 (b) local fault at F5.

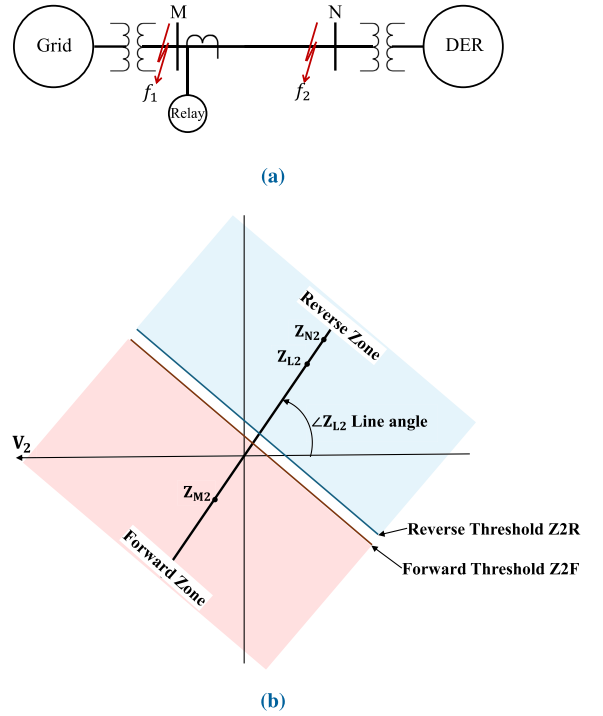
Negative sequence elements have a restraint factor of  $|I_2|/|I_1|$ , set typically between 0.1 to 0.2. This serves as a supervision to ensure the security of the element against normal imbalance conditions in the network. The resulting  $|I_2|/|I_1|$  from the low  $V_2$  (due to the remote B-C-G fault at F9) is 0.167, which can impact the reliability of the relay as backup for detecting the fault. As can be seen in Fig. 5a, the low remote fault  $V_2$ , potentially puts this restraint factor within the setting range. A similar fault nearer to the relay, at F5, however produces a  $|V_2|$  of 31.9Vsec, and  $I_2$  above the possible restraint factor.

## 2) IMPEDANCE-BASED NEGATIVE SEQUENCE DIRECTIONAL ELEMENT

An impedance-based approach to determining the fault direction via negative sequence is implemented in [43] to deal with the challenge identified with torque-product approach,  $T32Q$ . A basic negative sequence impedance ( $Z_2$ ) element is formulated as

$$Z_2 = \frac{\text{Re}[V_2 \cdot (1/\angle Z_{1\text{ANG}} \cdot I_1)^*]}{|I_2|^2} \quad (7)$$

This approach considers the fact that the  $V_2$  developed for forward faults is always negative. Thus, the negative sequence impedance measured for such a fault will be negative. Consider, for instance, the basic network shown in Fig. 6a. For a the forward fault  $f_2$ , the negative sequence impedance measured by the relay will be the impedance behind the relay ( $-Z_{M2}$ ). Conversely, for the reverse fault  $f_1$ , the measured impedance will be the sum of the impedances of the line and DER,  $Z_{L2}$  and  $Z_{N2}$ . These opposite impedances for forward and reverse zones provide the zones of protection



**FIGURE 6.** (a) Basic network. (b) Negative sequence impedance direction determination.

for the relay, as shown on Fig. 6b. To determine the direction, the impedance measured by the relay is compared with the determined thresholds for forward and reverse faults,  $Z_2^F$  and  $Z_2^R$  respectively, as shown in Fig. 6b.

The major challenge of this prevailing method is the unconventional nature of the negative-sequence network of IBR-dominated systems. Not only do such networks experience low  $|V_2|$ , but also they may not follow the assumed inductive nature used in formulating such methods [44]. Modifications to negative sequence approaches must therefore take into consideration the nature of  $I_2$  injection from IBRs [45]. As discussed in [16], the injection of reactive  $I_2$  by IBR controls during unbalanced faults, for instance as proposed in [46] or as outlined in the German grid code [37], provides the means to develop modern negative sequence directional methods.

Further analysis of the performance of this element is presented in Section VI.

## C. ZERO SEQUENCE DIRECTIONAL ELEMENT

Since zero-sequence quantities are present in significant amount, they are used to provide directional determination specifically for ground faults. Just as with positive and negative sequence methods, a torque-product method was originally implemented for zero-sequence. Two types of zero sequence directional protection are generally implemented based on the quantity selected as polarizing quantity. The current-polarized element, T32I, uses the neutral current from

a local transformer as polarizing quantity and is given by (8)

$$T32I = |I_{pol}| \cdot |I_0| \cdot \cos[\angle I_{pol} - \angle I_0] \quad (8)$$

where  $I_{pol}$  represents the local transformer neutral current and  $I_0$  is the zero-sequence quantity. The product is positive for forward faults and negative for reverse faults. This element relies on a consistent direction of  $I_{pol}$  under all circumstances and this is dependent on the transformer grounding configuration used.

The voltage-polarized element, T32V, uses the zero sequence voltage as polarizing quantity and is given by (9)

$$T32V = |V_0| \cdot |I_0| \cdot \cos[\angle -V_0 - (\angle I_0 + \angle Z_{L0})] \quad (9)$$

where  $Z_{L0}$  is the zero sequence impedance angle of the protected line and  $V_0$  is the zero sequence quantity, the polarizing quantity.

As has been discussed for T32P and T32Q, the elements in (8) and (9) are impacted by the presence of significantly reduced quantities due, for instance, high impedance faults. Moreover, since IBRs do not provide zero sequence, these elements are impacted by the interfacing transformer configuration.

#### D. SUMMARY OF DIRECTIONAL PROTECTION CHALLENGES

The effectiveness of traditional directional protection methods is impacted by different unique characteristics of the modern distribution grid as extensively discussed in the previous subsections. Overall, these challenges can be summarized as follows:

- 1) Traditional DOCR methods that rely on initial over-current detection of fault incidence are fundamentally impacted by the low fault current levels in islanded networks;
- 2) Fault currents that approach loading levels impact detection of unbalanced faults when positive sequence detection methods are employed;
- 3) Low  $V_2$  and  $I_2$  magnitudes during faults in islanded networks can render the angle of the quantities unreliable for various protection algorithms;
- 4) Zero-sequence methods rely on unique interfacing transformer grounding since IBRs generally do not provide zero sequence currents.

#### IV. PROPOSED PROTECTION METHOD

As demonstrated in Section III, the influence of IBR integration on bi-directionality and fault signal magnitudes in microgrids can adversely impact the reliability of conventional directional elements. As a solution, the authors employed superimposed or incremental quantities as a way to improve the sensitivity and selectivity of fault detection.

Consider, for instance a network with pre-fault phase currents  $I_A$ ,  $I_B$  and  $I_C$ , and, similarly, steady-state post-fault quantities  $I'_A$ ,  $I'_B$  and  $I'_C$ . Using the principle of superposition on the pre- and post-fault networks, we obtain a pure fault

network, with delta quantities,  $\Delta I_A$ ,  $\Delta I_B$ , and  $\Delta I_C$ , between the pre- and post-fault quantities. These quantities eliminate loading conditions in fault quantities in order to obtain only fault-driven values for detection. This applies also to voltage quantities.

The developed protection method introduced by the authors in [47] and evaluated in this paper, employs the use of superimposed negative sequence admittance quantities. This element is employed as a non-communication-based method to detect unbalanced faults which constitute about 95% of distribution system faults. It is based on obtained superimposed phase quantities. Symmetrical components  $I_0$ ,  $I_1$ , and  $I_2$  (zero, positive, and negative sequence currents, respectively) are obtained from the phase quantities as shown in (10)

$$\begin{bmatrix} I_0 \\ I_1 \\ I_2 \end{bmatrix} = \frac{1}{3} \begin{bmatrix} 1 & 1 & 1 \\ 1 & a & a^2 \\ 1 & a^2 & a \end{bmatrix} \begin{bmatrix} I_A \\ I_B \\ I_C \end{bmatrix} \quad (10)$$

$I_2$ , for this purpose can be stored by the numerical relay as a memory quantity,  $I_{2m}$ .

A 1 cycle post-fault values of the phase currents can also be used to obtain a memory value of negative sequence current,  $I_2$ , as shown in 11. Modern numerical relays are able to store such memory quantities for a few cycles at a time.

$$I_2 = \frac{1}{3}(I'_A + a^2 I'_B + a I'_C) \quad (11)$$

The superimposed negative sequence currents can thus be obtained as:

$$\begin{aligned} \Delta I_2 &= I_2 - I_{2m} \\ &= \frac{1}{3}[(I'_A + a^2 I'_B + a I'_C) - (I_A + a^2 I_B + a I_C)] \\ &= \frac{1}{3}[(I'_A - I_A) + a^2(I'_B - I_B) + a(I'_C - I_C)] \\ &= \frac{1}{3}[\Delta I_A + a^2 \Delta I_B + a \Delta I_C] \end{aligned} \quad (12)$$

This is repeated to obtain  $V_2$  and  $V_{2m}$  for the negative sequence voltages. The negative sequence admittance is then calculated as:

$$\Delta Y_2 = \frac{I_2 - I_{2m}}{V_2 - V_{2m}} \quad (13)$$

where  $\Delta Y_2$  represent the superimposed negative sequence admittance. In this case, 1 cycle is used as differentiator between the fault quantities and the memory quantities. The value obtained is thus a 1 cycle comparison between the change in current and the change in voltage.

Since  $\Delta Y_2$  is obtained as a complex quantity, the magnitude of the complex value must exceed a predetermined threshold for fault incidence to be detected. It is known that the measured  $Z_2$  by a relay is dependent on the  $Z_2$  of the source behind the relay. This provides a way to determine the minimum  $|\Delta Y_2|$  threshold for the relay setting as a proportion of the inverse impedance ( $1/Z_2$ ) of the source behind the relay. The  $Z_2$  of inverters are known not to stay constant. However,

the superimposed quantity allows for the incremental change of this impedance (or admittance in this case) to be captured within 1 cycle of a fault incident. This represents a fast extraction of the fault component preceding changes in the impedance.

The proof of concept and analysis provided in [47] shows that the phase angle of  $\Delta Y_2$  can be used to determine the direction of the fault. Since the argument of  $\Delta Y_2$  is dependent on the angle of  $1/\Delta Z_2$  behind the relay for a forward fault. This would result in a  $90^\circ$  for a perfectly inductive network. However consideration is given to less inductive conditions in this case by introducing a compensation factor. as  $X/R$  ratio increases with increase in resistive components, the compensation factor adjusts this angle anticlockwise. The forward zone criterion is given as

$$0^\circ + \phi < \arg\{\Delta Y_2\} < 180^\circ + \phi \quad (14)$$

whereas the reverse zone is given by

$$180^\circ + \phi < \arg\{\Delta Y_2\} < 360^\circ + \phi \quad (15)$$

where  $\phi$  is an angle that compensates for the specific protected line as well as measurement errors in the relaying process.

In order to improve the security of the system, the typical positive sequence restraint factor of  $|I_2|/|I_1|$  used in negative sequence based elements is implemented for this approach. Also, superimposed quantities are inherently short-lived based on a 1 cycle window. A latch-bit can therefore be implemented in microprocessor based relays to hold the superimposed quantities obtained based on restraint factor evaluation, and released following the completion of the fault detection and direction algorithm. A summary of the general fault detection logic is presented in Fig. 7.

## V. HARDWARE-IN-THE-LOOP (HIL) TEST SETUP

This section discusses the testbed used to evaluate the performance of the proposed method. The test microgrid is simulated in real-time using the multi-core OPAL-RT real digital simulator platform together with SEL-411L protection relays and Siemens Ruggedcom RS900G managed Ethernet switch. These hardware devices are configured in a local area network managed by the Ethernet switch. Fig. 8 depicts the basic architecture of the testbed.

### A. REAL TIME DIGITAL SIMULATOR

The OPAL-RT processing unit, OP5600, has multiple cores for executing real-time electromagnetic transient simulations. In this test, Core 1 is assigned to solving the modified IEEE 13-node test feeder, shown in Fig. 1 using the ARTEMiS solver. This includes the network components and the control system implemented. The test feeder is modeled using the MATLAB/Simulink environment, through RT-LAB which is the software interface for OPAL-RT. This model is solved based on a fixed time step of  $25\mu s$ . The OP5600 is also equipped with communication ports and analog input/output

(I/O) channels which are used to interface with the external hardware. Core 2 is assigned to the external hardware interface.

### B. PROTECTION RELAYS

The SEL-411L relay [48] is integrated into the real-time simulator to implement the  $\Delta Y_2$  protection method. Low-voltage analog signals are supplied by the OPAL-RT's OP5330 analog output board to the relay via an Omicron CMS Amplifier. The low-voltage signals are amplified as voltage and current inputs for the relay, similar to CT and VT secondary quantities. The relays are integrated at R633, R632 and R671 of Fig. 1.

For a voltage or current signal,  $x$ , (16) provides the general concept for obtaining superimposed sample from the relay.

$$\Delta x_k = x_k - x_{k+N} \quad (16)$$

$\Delta x_k$  is the superimposed quantity,  $x_k$  is the present voltage or current sample, and  $N$  is the total number of samples per cycle. Thus to be able to obtain a 1 cycle difference, the present sample must be stored for the number of samples it takes to obtain a time of 1 cycle, that is,  $k + N$ .

The relay settings are programmed using the SEL-5030 acSELeRator QuickSET software. The proposed protection method is implemented in the relays using the protection free-form SELogic control equation capability of the acSELeRator QuickSET software. It provides math variables together with Boolean and other operators through the free-form logic control. This allows for the custom protection logic to be developed. In order to delay the current and voltage signals by 1 cycle and obtain a superimposed quantity for the  $\Delta Y_2$ , the quantity is repeated with nine math variables (PMV01 to PMV09). Appendix B shows how the algorithm is created to obtain the  $\Delta I_2$  and  $\Delta V_2$  quantities.

The relays also feature the inherently available Z2 directional protection element, in order to compare the results with the  $\Delta Y_2$  method. The designated trip contact of the relay is connected to the OPAL-RT's OP5353 digital input board to provide signal for breaker operation. Breakers are modeled as part of the test feeder in MATLAB/Simulink.

### C. COMMUNICATION NETWORK

OP5600 can integrate different types of external hardware over local area network (LAN) using standard communication protocols. RT-LAB's interface is used to establish the DNP3 communication protocol between the OP5600 target and the relays. The I/O interface of RT-LAB assigns the OP5600 as the DNP3 Client interface to receive analog data from the relay operation (DNP3 Server). These quantities can be viewed from the connected workstation. The Ruggedcom RS900G Ethernet switch is used to manage this LAN.

## VI. RESULTS AND DISCUSSION

The proposed superimposed negative sequence admittance method was tested for various fault locations in the



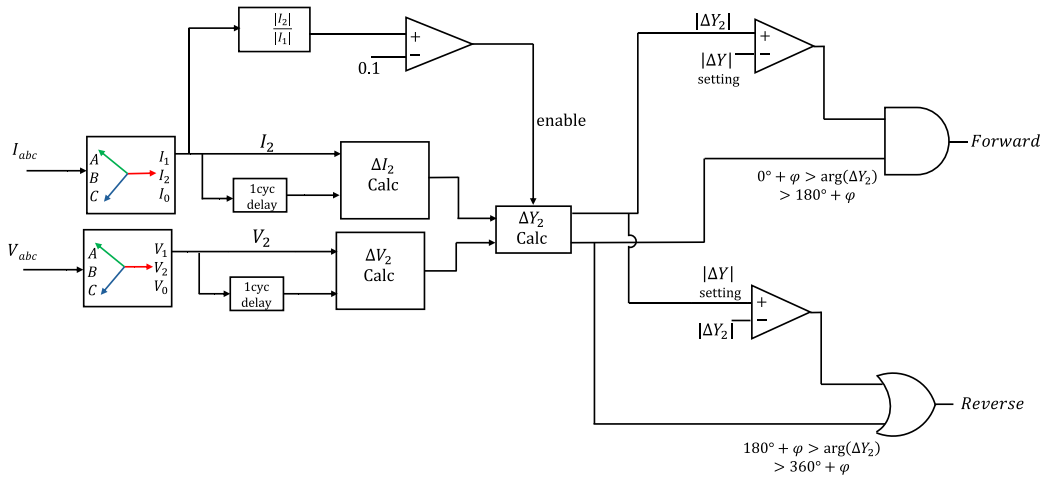


FIGURE 7. Relay logic for  $\Delta Y_2$ .

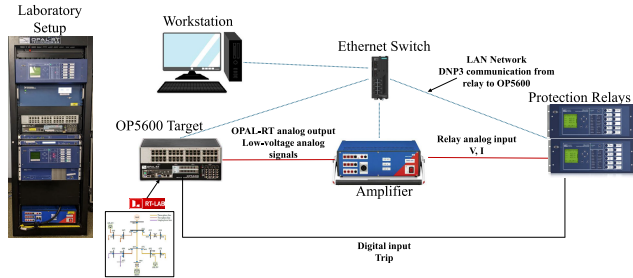


FIGURE 8. Layout of the Hardware-in-the-Loop (HIL) testbed.

distribution system shown in Fig. 1. The evaluation was carried out for both grid-connected and islanded mode. The multiple setting group capability of the relays is used to implement different settings for the two modes due to the differences in power flows. A minimum fault resistance of  $5\Omega$  is assumed in each case. Since the negative sequence impedance  $|Z_2|$  measured by the relay is dependent on the impedance of the source behind the relay, the minimum setting  $|Y_{set}|$  (i.e  $1/|Z_2|$ ) is determined by considering the minimum permissible voltage for a normal condition or a reverse fault.  $|\Delta Y_2|$  should exceed this value to register a forward fault. A minimum allowable grid voltage of 0.9pu is thus used in this case for the setting calculations. From the foregoing analysis, a range of 0.4 to 0.80pu is obtained for various relays under study. A conservative  $|Y_{set}|$  value of 1pu is set for the relays for simplicity. This provides sufficient margin for the relays to remain both secure and dependable.

## A. GRID-CONNECTED MODE

### 1) RELAY R2

The performance of the  $\Delta Y_2$  method in the grid-connected mode is first tested on relay R2. A 2LG fault (B-C-G) is placed at F2. This is in the forward direction of the relay. This

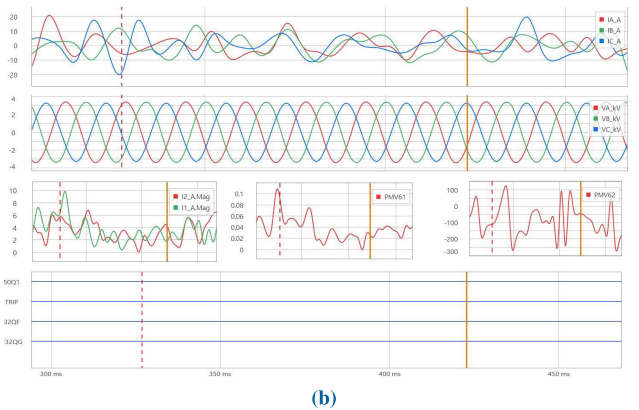
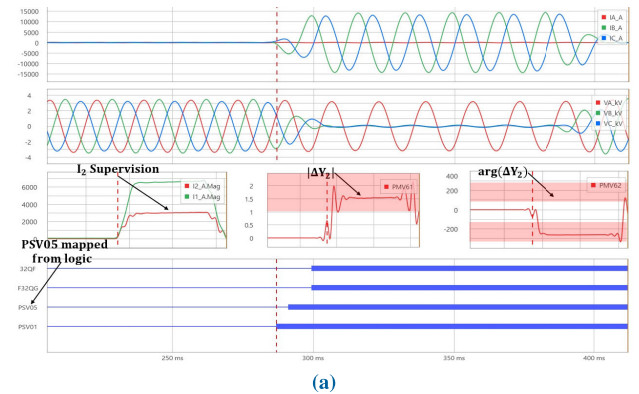


FIGURE 9. Relay R2 event for a (a) forward fault (b) reverse fault.

relay functions in a radial section and, apart from the source behind it, will also be impacted mainly by the load. Using the superimposed quantity method eliminates the impact of the load on the fault detection. A line impedance angle magnitude setting of  $89.85^\circ$  is used together with the AUTO2 directional control setting in the SEL relay for the internal detection method.

Fig. 9a shows a record of the fault from SEL SynchroWAVE Event. Three custom analog charts are created from the math variables to show the result from the relay operation of the proposed  $\Delta Y_2$  element programmed in the relay logic. These provide visualization for the  $|I_1|/|I_2|$  starting criterion, the  $|\Delta Y_2|$  value, and the  $\arg(\Delta Y_2)$  value. As can be seen from the  $|\Delta Y_2|$  chart, after the initial fault transient, a value of 1.529pu is obtained, beyond the minimum setting. Based on the analysis by the authors in [47], a compensation angle of  $45^\circ$  is used. Red shaded regions are included in the  $\Delta Y_2$  charts to show the forward fault range. The  $\arg(\Delta Y_2)$  obtained for this scenario is  $-261.6^\circ$ . PSV05 is the wordbit used to represent the output of the element.

As shown in the digital chart of the event report, it is asserted after 2.5 cycles, indicating that conditions for the fault are met and the relay correctly identifies the fault as such. The SEL-411L relay uses the negative sequence impedance (Z2) element discussed in Section III-B2 as its internal method for determining the direction for unbalanced faults. The analog chart of Fig. 9a shows the outputs of the internal Z2 method, F32QG and 32QF, which are also asserted for a forward fault in this scenario.

The performance of R2 is also examined by considering a similar B-C-G fault at F3. This fault is reverse to the relay. The event record in Fig. 9b shows a manually triggered event after the fault, as the relay accurately did not recognize the reverse fault and as such did not automatically register the fault. This is revealed in the analog chart as neither the PSV05 assigned for the  $\Delta Y_2$  nor the F32QG and 32QF outputs for the internal Z2 element were asserted. The analog charts (with the exception of the voltage chart) represent noise captured after the opening of the upstream breaker.

## 2) RELAYS R632 AND R671

A similar test is repeated to evaluate the method in relays R632 and R671 based on a fault on the line between bus 632 and 671. The relays are positioned to ensure that the line is in their respective primary zone of protection. A fault on the line must be detected and tripped by both relays to isolate the fault. For this scenario, injection of  $I_2$  by the BESS is implemented to reflect recommendations in [46]. More so, this case provides a test for the bi-directional flow of fault current which is a major challenge in microgrids.

A B-C-G fault at F2 is considered. R632 is closer to the grid side and the negative sequence impedance measurement is dependent on the source behind it. Since the grid is a stronger source and the more significant contributor of fault current, negative sequence measurement is sufficient to provide fault detection for both the internal Z2 and the  $\Delta Y_2$  methods. The Fig. 10a shows a  $|\Delta Y_2|$  value of 1.519pu, above the setting. Similarly, based on a compensation angle of  $45^\circ$ , the relay accurately determines the fault in the forward direction at  $93.9^\circ$ . The digital chart also shows assertion by all relevant outputs for forward fault detection. Similar to R2, the fault

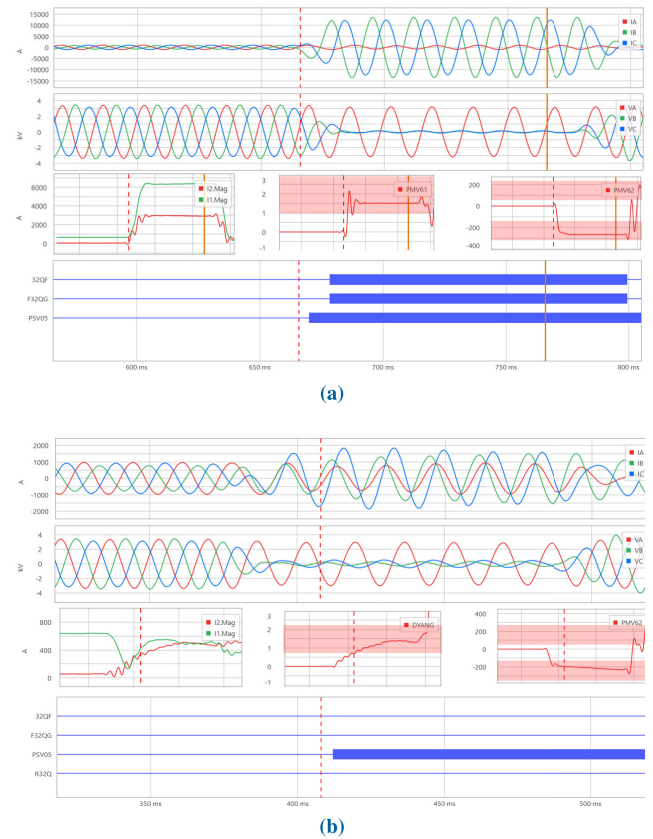


FIGURE 10. Response to B-C-G fault at F5 by (a) R632 (b) R671.

detection method is asserted at a response time of about 2.5 cycles.

Relay R671, however, depends on the  $I_2$  injection by the BESS. The load imbalance in the network can thus significantly impact the contribution to the fault, and subsequently impact the operation of the R671 based on negative sequence quantities. A significantly low  $I_2$  renders the current angle unreliable for the formulation in (7). As can be seen from Fig. 10b, a limited fault current is contributed by the BESS. More so, the  $I_2$  of 504A (compared with 2.9kA at R632) is significantly lower. The outputs for the internal Z2 element, F32QG and 32QF, fail to detect the fault. This reveals that the outputs of the internal Z2 element result in misoperation due to the unreliability of the negative sequence quantities. This challenge due to the load does not affect the  $\Delta Y_2$  as the fault components devoid of load influence are used. R671 thus correctly determines the forward fault with  $|\Delta Y_2|$  and  $\arg(\Delta Y_2)$  at 1.19pu and  $-210^\circ$ , respectively, at the time of tripping after 6 cycles.

Another approach to detecting and isolating this fault is to implement a sequential operation involving initial detection and trip by R632, reconfiguration of the network to islanded mode and final detection and trip by R671. In this scenario, an adaptive protection scheme may be implemented using multiple setting groups in the relay. Settings for the islanded mode, after reconfiguration, are then applied to operate R671.

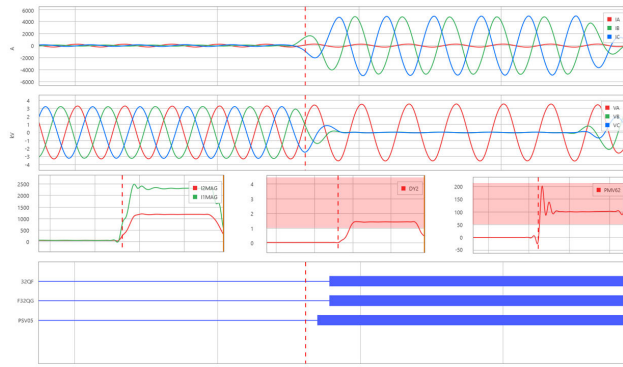


FIGURE 11. Relay R692 event for a fault at F8.

The key factor in this case is the speed of operation of the control and protection schemes. The operation of R671 in islanded mode is discussed in Section VI-B.

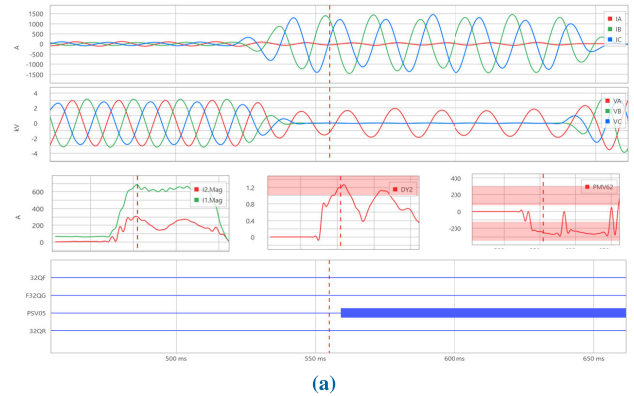
### 3) RELAY R692

Further testing was conducted at bus 692 to validate the method on the three-phase line between section 671 to 675. One unique feature of this line in the modified network is the presence of the 3-phase PV. It operates in the GFL-mode and only increases its positive sequence current contribution to the rated maximum in the event of a fault. Thus, this source does not impact the operation of both the  $|\Delta Y_2|$  method or the existing Z2 directional method which depend on negative sequence quantities. It can be seen from Fig. 11 that both F32G and PSV05 assert for the forward fault at 692, with the latter asserting 1 cycle faster in this case (when no time delay setting is included). The novel method accurately detects the fault, for a magnitude of 1.58 and an angle of  $97.4^\circ$ , both within the forward fault zone. Another important feature of this branch is the higher level of imbalance due to the load in the test system. It is important to adjust the starting criterion  $|I_2|/|I_1|$  appropriately for this relay in order to prevent nuisance trips due to imbalance in normal load conditions.

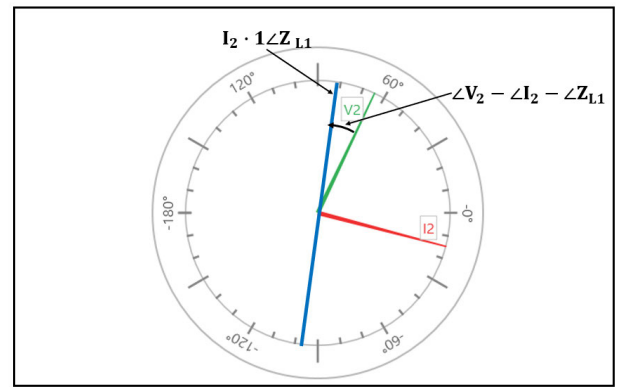
Table 1 presents response of the relays in the network to faults in the grid-connected mode. Results for single-line-to-ground (SLG), double-line-to-ground (2LG), and line-to-line (2LG) faults are presented on the three-phase lines in the network. An observation can be made about the  $\Delta Y_2$  measurement values of R633 which serves as a remote relay to protect the section 632-633. Due to the low negative sequence impedance behind it, a relatively higher  $I_2$  value will be recorded by the relay based on the current divider rule, leading to a relatively higher  $|\Delta Y_2|$  measurement. However, the relay logic detects the fault correctly based on  $\arg(\Delta Y_2)$ .

### B. ISLANDED MODE

Two challenges arise in the islanded mode for conventional directional overcurrent fault detection methods. Firstly, the lower short-circuit currents impact the overcurrent fault



(a)



(b)

FIGURE 12. (a) R2 response to forward B-C-G fault at F2 in islanded mode (b) Misoperation of Z2 element in R2 for fault at F2 based on  $I_2$  and  $V_2$  measured.

detection approach. Secondly, sequence quantity injection levels must be significant to allow for sequence-based directional determination. It is, therefore important for GFM control schemes to be designed to emulate conventional negative sequence contribution within current-limitation and grid-code boundaries. Due to the lower negative sequence injection, a  $|\Delta Y_2|$  setting of 0.8 is used in this mode. The  $\arg(\Delta Y_2)$  setting is maintained as in the grid-connected mode for simplicity.

### 1) RELAY R2

Fig. 12a shows the result of a B-C-G fault in the forward direction of R2, at F2. As mentioned previously, a significant reduction in faulted phase current is observed from 14.1kA peak current in the grid-connected mode to 1.4kA in this mode. This, compounded with the lower negative sequence contribution from the BESS, poses a challenge to the internal directional element of the relay. The AUTO2 setting of this element is used to account for the higher negative sequence impedance of the IBR and potential inaccuracies in the line impedance determination.

Fig. 12b shows the angle between  $V_2$  and  $(I_2 \cdot 1/Z_{L1})$ . The image superimposes the  $(I_2 \cdot 1/Z_{L1})$  line on the event record of  $V_2$  and  $I_2$  obtained from the SynchroWAVE Event software.

**TABLE 1.** Fault direction decision using  $\Delta Y_2$  on three-phase lines of Fig. 1 in the grid-connected mode.

| SLG Fault (A-G) |                |                    |          |                |                    |          |                |                    |          |
|-----------------|----------------|--------------------|----------|----------------|--------------------|----------|----------------|--------------------|----------|
|                 | F1             |                    |          | F2             |                    |          | F3             |                    |          |
| Relay           | $ \Delta Y_2 $ | $\arg(\Delta Y_2)$ | Dir      | $ \Delta Y_2 $ | $\arg(\Delta Y_2)$ | Dir      | $ \Delta Y_2 $ | $\arg(\Delta Y_2)$ | Dir      |
| R634            | 1.405          | 131.1              | <i>F</i> | 0.026          | -17.7              | <i>R</i> | 0.032          | -33.4              | <i>R</i> |
| R633            | 0.949          | -72.1              | <i>R</i> | 1.131          | 147.0              | <i>F</i> | 1.134          | 147.1              | <i>F</i> |
| R2              | 1.514          | 104.2              | <i>F</i> | 1.521          | 111.4              | <i>F</i> | 0.017          | -33.5              | <i>R</i> |
| REC1            | 3.198          | 172.2              | <i>F</i> | 3.257          | 172.0              | <i>F</i> | 3.310          | 171.7              | <i>F</i> |
|                 | F3             |                    |          | F5             |                    |          | F6             |                    |          |
| R632            | 0.001          | 7.8                | <i>R</i> | 1.601          | 153.5              | <i>F</i> | 1.613          | 153.6              | <i>F</i> |
| R671            | 1.307          | 150.4              | <i>F</i> | 1.475          | 150.8              | <i>F</i> | 0.882          | -70.9              | <i>R</i> |
| R680            | 1.003          | -150.2             | <i>F</i> | 1.003          | -162.7             | <i>F</i> | 1.031          | -165.8             | <i>F</i> |
|                 | F7             |                    |          | F8             |                    |          | F9             |                    |          |
| R675            | 1.257          | 114.6              | <i>F</i> | 0.053          | -26.4              | <i>R</i> | 0.052          | -25.8              | <i>R</i> |
| R692            | 1.235          | 111.3              | <i>F</i> | 1.248          | 111.7              | <i>F</i> | 0.083          | -59.5              | <i>R</i> |
| R3              | 1.781          | 111.6              | <i>F</i> | 1.783          | 112.1              | <i>F</i> | 1.780          | 115.0              | <i>F</i> |

| 2LG Fault (B-C-G) |                |                    |          |                |                    |          |                |                    |          |
|-------------------|----------------|--------------------|----------|----------------|--------------------|----------|----------------|--------------------|----------|
|                   | F1             |                    |          | F2             |                    |          | F3             |                    |          |
| Relay             | $ \Delta Y_2 $ | $\arg(\Delta Y_2)$ | Dir      | $ \Delta Y_2 $ | $\arg(\Delta Y_2)$ | Dir      | $ \Delta Y_2 $ | $\arg(\Delta Y_2)$ | Dir      |
| R634              | 1.227          | 126.5              | <i>F</i> | 0.022          | -37.2              | <i>R</i> | 0.021          | -38.0              | <i>R</i> |
| R633              | 0.801          | -73.9              | <i>R</i> | 1.154          | 139.5              | <i>F</i> | 1.150          | 138.2              | <i>F</i> |
| R2                | 1.518          | 102.4              | <i>F</i> | 1.529          | 98.4               | <i>F</i> | 0.014          | -36.9              | <i>R</i> |
| REC1              | 3.300          | 171.1              | <i>F</i> | 3.285          | 172.4              | <i>F</i> | 3.269          | 172.0              | <i>F</i> |
|                   | F3             |                    |          | F5             |                    |          | F6             |                    |          |
| R632              | 0.001          | 7.9                | <i>R</i> | 1.519          | 93.9               | <i>F</i> | 1.503          | 93.8               | <i>F</i> |
| R671              | 1.085          | 149.4              | <i>F</i> | 1.190          | 150.3              | <i>F</i> | 0.821          | -71.3              | <i>R</i> |
| R680              | 1.032          | -146.2             | <i>F</i> | 1.048          | -151.6             | <i>F</i> | 1.168          | -166.3             | <i>F</i> |
|                   | F7             |                    |          | F8             |                    |          | F9             |                    |          |
| R675              | 1.541          | 112.6              | <i>F</i> | 0.028          | 32.1               | <i>R</i> | 0.029          | 100.4              | <i>R</i> |
| R692              | 1.584          | 95.8               | <i>F</i> | 1.578          | 97.5               | <i>F</i> | 0.034          | 35.5               | <i>R</i> |
| R3                | 1.593          | 98.0               | <i>F</i> | 1.586          | 96.9               | <i>F</i> | 1.567          | 97.6               | <i>F</i> |

| 2L Fault (B-C) |                |                    |          |                |                    |          |                |                    |          |
|----------------|----------------|--------------------|----------|----------------|--------------------|----------|----------------|--------------------|----------|
|                | F1             |                    |          | F2             |                    |          | F3             |                    |          |
| Relay          | $ \Delta Y_2 $ | $\arg(\Delta Y_2)$ | Dir      | $ \Delta Y_2 $ | $\arg(\Delta Y_2)$ | Dir      | $ \Delta Y_2 $ | $\arg(\Delta Y_2)$ | Dir      |
| R634           | 1.222          | 127.2              | <i>F</i> | 0.022          | -37.2              | <i>R</i> | 0.023          | -38.5              | <i>R</i> |
| R633           | 0.811          | -73.2              | <i>R</i> | 1.122          | 144.7              | <i>F</i> | 1.230          | 142.9              | <i>F</i> |
| R2             | 1.519          | 103.0              | <i>F</i> | 1.520          | 99.2               | <i>F</i> | 0.012          | -37.9              | <i>R</i> |
| REC1           | 3.270          | 171.7              | <i>F</i> | 3.283          | 172.3              | <i>F</i> | 3.275          | 172.0              | <i>F</i> |
|                | F3             |                    |          | F5             |                    |          | F6             |                    |          |
| R632           | 0.001          | 7.4                | <i>R</i> | 1.488          | 168.6              | <i>F</i> | 1.437          | 169.4              | <i>F</i> |
| R671           | 1.115          | 147.6              | <i>F</i> | 1.186          | 147.3              | <i>F</i> | 0.858          | -81.7              | <i>R</i> |
| R680           | 1.021          | 71.9               | <i>F</i> | 1.025          | 72.8               | <i>F</i> | 1.094          | 50.03              | <i>F</i> |
|                | F7             |                    |          | F8             |                    |          | F9             |                    |          |
| R675           | 1.594          | 104.8              | <i>F</i> | 0.016          | -1.8               | <i>R</i> | 0.016          | -1.9               | <i>R</i> |
| R692           | 1.694          | 102.6              | <i>F</i> | 1.654          | 102.3              | <i>F</i> | 0.021          | 38.8               | <i>R</i> |
| R3             | 1.729          | 172.3              | <i>F</i> | 1.796          | 172.2              | <i>F</i> | 1.782          | 173.2              | <i>F</i> |

*F* - Forward direction; *R* - Reverse direction

The angle leads to a positive cosine. Based on calculation of Z2 settings in [48] and provided in Appendix A, a Z2 value of 2.83 is obtained. For a forward fault, the

formulation of the Z2 element requires this value to be lower than Z2F threshold, as shown in Fig. 6b. The Z2F threshold (Fig. 6) is pre-determined, however, to be  $-0.94$ ,



**TABLE 2.** Fault direction decision using  $\Delta Y_2$  on three-phase lines of Fig.1 in islanded mode.

| SLG Fault (A-G) |                |                    |     |                |                    |     |                |                    |     |
|-----------------|----------------|--------------------|-----|----------------|--------------------|-----|----------------|--------------------|-----|
|                 | F1             |                    |     | F2             |                    |     | F3             |                    |     |
| Relay           | $ \Delta Y_2 $ | $\arg(\Delta Y_2)$ | Dir | $ \Delta Y_2 $ | $\arg(\Delta Y_2)$ | Dir | $ \Delta Y_2 $ | $\arg(\Delta Y_2)$ | Dir |
| R634            | 1.447          | -176.5             | F   | 0.084          | -35.7              | R   | 0.076          | -34.2              | R   |
| R633            | 0.741          | 4.4                | R   | 0.981          | 144.2              | F   | 0.976          | 149.7              | F   |
| R2              | 1.422          | -172               | F   | 1.673          | 169.1              | F   | 0.018          | -31.4              | R   |
|                 | F3             |                    |     | F5             |                    |     | F6             |                    |     |
| R671            | 1.298          | 151.5              | F   | 1.403          | 164.1              | F   | 0.865          | -15.6              | R   |
| R680            | 1.015          | -178.1             | F   | 1.012          | -167.5             | F   | 1.005          | 158.1              | F   |
|                 | F7             |                    |     | F8             |                    |     | F9             |                    |     |
| R675            | 1.298          | 151.5              | F   | 0.042          | -52.6              | R   | 0.041          | -47.2              | R   |
| R692            | 1.376          | 148.9              | F   | 1.345          | 149.1              | F   | 0.032          | -28.9              | R   |
| R3              | 1.377          | 153.1              | F   | 1.343          | 152.8              | F   | 1.339          | 151.2              | F   |

| 2LG Fault (B-C-G) |                |                    |     |                |                    |     |                |                    |     |
|-------------------|----------------|--------------------|-----|----------------|--------------------|-----|----------------|--------------------|-----|
|                   | F1             |                    |     | F2             |                    |     | F3             |                    |     |
| Relay             | $ \Delta Y_2 $ | $\arg(\Delta Y_2)$ | Dir | $ \Delta Y_2 $ | $\arg(\Delta Y_2)$ | Dir | $ \Delta Y_2 $ | $\arg(\Delta Y_2)$ | Dir |
| R634              | 1.221          | 140.0              | F   | 0.057          | -20.9              | R   | 0.056          | -20.2              | R   |
| R633              | 0.317          | -45.9              | R   | 0.903          | 160.3              | F   | 0.907          | 158.9              | F   |
| R2                | 1.320          | 134.8              | F   | 1.150          | 139.3              | F   | 0.013          | -21.1              | R   |
|                   | F3             |                    |     | F5             |                    |     | F6             |                    |     |
| R671              | 1.163          | 154.7              | F   | 1.141          | 158.4              | F   | 0.081          | -50.4              | R   |
| R680              | 1.003          | 116.8              | F   | 1.004          | 119.4              | F   | 1.002          | 139.1              | F   |
|                   | F7             |                    |     | F8             |                    |     | F9             |                    |     |
| R675              | 1.196          | 120.3              | F   | 0.061          | 26.8               | R   | 0.054          | 28.9               | R   |
| R692              | 1.105          | 115.8              | F   | 1.092          | 122.2              | F   | 0.060          | 41.5               | R   |
| R3                | 1.097          | 106.5              | F   | 1.094          | 106.1              | F   | 1.097          | 139.8              | F   |

| 2L Fault (B-C) |                |                    |     |                |                    |     |                |                    |     |
|----------------|----------------|--------------------|-----|----------------|--------------------|-----|----------------|--------------------|-----|
|                | F1             |                    |     | F2             |                    |     | F3             |                    |     |
| Relay          | $ \Delta Y_2 $ | $\arg(\Delta Y_2)$ | Dir | $ \Delta Y_2 $ | $\arg(\Delta Y_2)$ | Dir | $ \Delta Y_2 $ | $\arg(\Delta Y_2)$ | Dir |
| R634           | 1.180          | 143.2              | F   | 0.051          | -28.9              | R   | 0.050          | -27.6              | R   |
| R633           | 0.329          | -51.7              | R   | 0.919          | 149.1              | F   | 0.918          | 147.9              | F   |
| R2             | 1.339          | 133.2              | F   | 1.303          | 141.3              | F   | 0.017          | -31.3              | R   |
|                | F3             |                    |     | F5             |                    |     | F6             |                    |     |
| R671           | 1.267          | 148.2              | F   | 1.261          | 144.4              | F   | 0.079          | -39.5              | R   |
| R680           | 1.006          | 134.4              | F   | 1.007          | 126.7              | F   | 1.006          | 122.3              | F   |
|                | F7             |                    |     | F8             |                    |     | F9             |                    |     |
| R675           | 1.229          | 135.4              | F   | 0.026          | 13.7               | R   | 0.022          | 13.1               | R   |
| R692           | 1.246          | 115.8              | F   | 1.244          | 122.2              | F   | 0.031          | 41.5               | R   |
| R3             | 1.239          | 133.8              | F   | 1.238          | 135.8              | F   | 1.254          | 132.1              | F   |

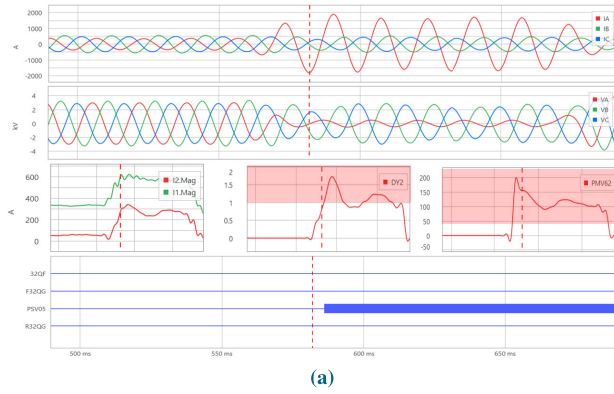
F - Forward Direction; R - Reverse Direction

hence the element misoperates as Z2 is beyond the Z2F threshold.

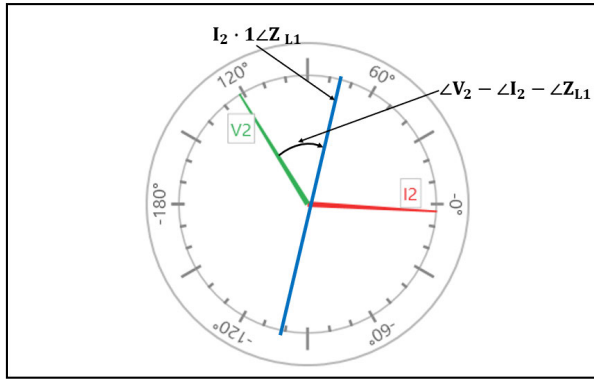
$\Delta Y_2$  on the other hand is able to accurately detect the incidence and occurrence of the fault in 2.5 cycles as in the grid-connected mode. It can be observed, however that apparent fluctuation in the the control action of the  $I_2$  injection, can impact the consistency in the measurements for  $|\Delta Y_2|$  and  $\arg(\Delta Y_2)$ . This is evident in the dips observed in both quantities.

## 2) RELAY R671

Similar to the operation of R2 in islanded mode, R671 also successfully detects a B-C-G fault at F5 in its forward direction using  $\Delta Y_2$ . As shown in Fig. 13a, a  $|\Delta Y_2|$  value of 1.141 is recorded at the time of trip at 2 cycles and this rises to a peak value of 1.731pu. Fluctuations in  $I_2$  injection causes the  $|\Delta Y_2|$  value to drop below the 0.8pu setting after 3.75cyc. This effect of the control action can potentially affect the use of the element as a time-delayed backup.



(a)



(b)

**FIGURE 13.** (a) R671 response to forward B-C-G fault at F5 in islanded mode (b) Misoperation of Z2 element for fault at F5 based on R671 on  $I_2$  and  $V_2$  measured.

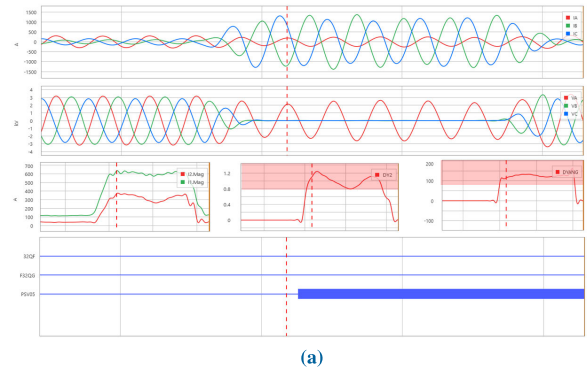
Fig. 13b shows that, just as in the case of R2, the  $Z_2$  angle developed between  $V_2$  and  $(I_2 \cdot 1/Z_{L1})$  is less than  $90^\circ$  resulting in forward misoperation. It is important to note that only the negative-sequence fault direction is recorded in this case. The SEL-411L relay has a variety of elements that are used concurrently to detect the fault incidence and direction.

### 3) RELAY R692

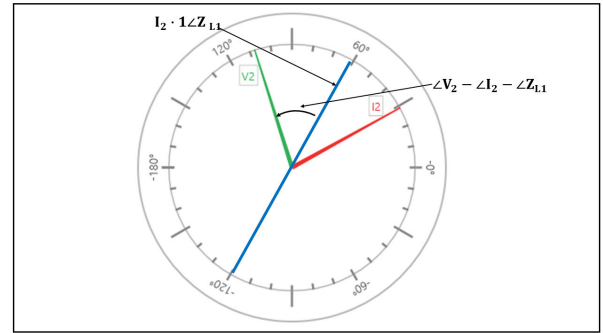
Fig. 14a shows the result of the B-C-G fault at Bus 692 in the islanded mode. Similar to other faulted areas in the islanded mode, the fault signature experiences instability challenges following fault incidence due to the IBRs and the  $I_2$  injection. This can be observed by the dip in fault signature after 1.5 cycles. As in the grid-connected mode, the higher level of imbalance due to normal load demands that the starting criterion be adjusted to prevent nuisance trips. The  $\Delta Y_2$  method successfully detected the fault via both the magnitude and direction at 1.5 cycles (instantaneous setting).

Fig. 14b shows the misoperation of the Z2 due to the angle of Z2 element obtained, as in the case of 671. For this fault however, the 50Q1 element (not shown) tripped for the fault.

Table 2 presents results of  $\Delta Y_2$  for other relays in the network in the islanded mode for SLG, 2LG and 2L faults.



(a)



(b)

**FIGURE 14.** (a) Relay R692 response to forward B-C-G fault at F8 in islanded mode (b) Misoperation of Z2 element in R692 for fault at F8 based on  $I_2$  and  $V_2$  measured.

## VII. CONCLUSION

This paper evaluates and demonstrates the potential issues with traditional directional protection methods used in conventional distribution system. It, therefore, sheds light on the specific challenges in developing protection mechanisms to address bi-directional fault currents in microgrids. With a focus on unbalanced faults, which constitute about 95% of distribution system faults, the negative sequence directional protection elements were considered.  $T32Q$  and Z2 elements were shown to be significantly impacted by the negative sequence characteristics of the IBRs. The low levels of negative sequence quantities that occur due to unbalanced loads in the network, have significant impact on available quantities for determining fault direction, especially in the islanded mode.

A superimposed sequence quantity approach is adopted to thus eliminate the impact of the load for fault detection, and rely on the fault component of the negative sequence quantity. The  $\Delta Y_2$  quantity is implemented in this case. The logic of this method can potentially be directly implemented in commercial relays with free-form logic control. A real-time evaluation of the method using the SEL-411L verifies it in both grid-connected and islanded modes. Since the negative-sequence contribution in the grid-connected mode is based on the main grid, it can be deduced that the method is also effective for the no IBR mode. It was however observed that, the effectiveness of this approach, particularly in the islanded mode, is dependent on the speed of the control

**TABLE 3.** Determining thresholds for Z2 element.

| Parameter | Value/Formula                                    | Description                                     |
|-----------|--------------------------------------------------|-------------------------------------------------|
| Z2        | $\frac{Re[V_2 - (I_2 Z_{L200} I_1)^*]}{ I_2 ^2}$ | Directional element calculation                 |
| Z2F       | -0.3                                             | Forward directional Z2 threshold. Default value |
| Z2R       | 0.3                                              | Reverse directional Z2 threshold. Default value |
| Z2FTH     | $0.75 \cdot Z2F - 0.25 \cdot  V_2 / I_2 $        | Calculated forward directional threshold        |
| Z2RTH     | $0.75 \cdot Z2R + 0.25 \cdot  V_2 / I_2 $        | Calculated reverse directional threshold        |

```

##-----I2 SUPERVISION-----
PMV22 := (L3I2FIM / 3.000000) / L1I2FIM #I2 SUPERVISION
PSV01 := PMV22 > 0.250000
#-----INCREMENTAL CURRENT-----
PMV09 := PMV08 #DELATED BY 1CYCLE
PMV08 := PMV07
PMV07 := PMV06
PMV06 := PMV05
PMV05 := PMV04
PMV04 := PMV03
PMV03 := PMV02
PMV02 := PMV01
PMV01 := 3I2A2NFR / 3.000000 # REAL PART NEGATIVE-SEQUENCE INSTANTANEOUS CURRENT, LINE
PMV10 := PMV01 - PMV09 #INCREMENTAL REAL PART
PMV19 := PMV18 #DELATED BY 1CYCLE
PMV18 := PMV17
PMV17 := PMV16
PMV16 := PMV15
PMV15 := PMV14
PMV14 := PMV13
PMV13 := PMV12
PMV12 := PMV11
PMV11 := 3I2A2NFI / 3.000000 # IMAG PART NEGATIVE-SEQUENCE INSTANTANEOUS CURRENT, LINE
PMV20 := PMV11 - PMV19 #INCREMENTAL IMAG PART33
#CALCULATING MAGNITUDE OF INCREMENTAL CURRENT
PMV21 := SQR(PMV20 * PMV20 + PMV10 * PMV10) #I2 MAGNITUDE
#-----INCREMENTAL VOLTAGE-----
PMV29 := PMV28 #DELATED BY 1CYCLE
PMV28 := PMV27
PMV27 := PMV26
PMV26 := PMV25
PMV25 := PMV24
PMV24 := PMV23
PMV23 := PMV22
PMV22 := PMV21
PMV21 := 3VA2YFR / 3.000000 # REAL PART NEGATIVE-SEQUENCE INSTANTANEOUS VOLTAGE MAGNITUDE, LINE
PMV40 := PMV31 - PMV29 #INCREMENTAL REAL PART
PMV49 := PMV48 #DELATED BY 1CYCLE
PMV48 := PMV47
PMV47 := PMV46
PMV46 := PMV45
PMV45 := PMV44
PMV44 := PMV43
PMV43 := PMV42
PMV42 := PMV41
PMV41 := 3VA2YFI / 3.000000 # IMAG PART NEGATIVE-SEQUENCE INSTANTANEOUS VOLTAGE, LINE
PMV50 := PMV41 - PMV49 #INCREMENTAL IMAG PART33
#CALCULATING MAGNITUDE OF INCREMENTAL VOLTAGE
PMV51 := SQR(PMV40 * PMV40 + PMV50 * PMV50) #V2 MAGNITUDE

```

**FIGURE 15.** User-defined algorithm to obtain superimposed quantities.

system in providing negative sequence injection to support the network during a fault. More so, further investigation must be carried out to address challenges with high impedance faults.

Future research will focus on defining setting rules for the new algorithm considering diverse network conditions, as is the case for conventional protection philosophies. Such investigation can consider empirically defining the impact of the IBR impedance on the setting ranges for the proposed  $\Delta Y_2$ -based method. Similarly, standardization in  $I_2$  injection, will provide a way to define generic relay setting rules in this regard.

## APPENDIX A

### SEL-411L DIRECTIONAL THRESHOLD CALCULATION

Table 3 shows different setting rules and calculations to determine the forward and reverse threshold for the Z2 element found internally in the SEL-411L relay.

## APPENDIX B

### ALGORITHM FOR OBTAINING SUPERIMPOSED QUANTITIES IN SEL-411L

A sample algorithm to extract superimposed negative sequence voltages and currents, is provided in Fig 15. Obtaining  $\Delta I_2$  and  $\Delta V_2$  are also demonstrated.

## ACKNOWLEDGMENT

The views expressed herein do not necessarily represent the views of the U.S. Department of Energy or the United States Government.

## REFERENCES

- [1] C. D'Adamo, C. Abbey, A. Baitech, A. Norsk, B. Matjila, B. Buchholz, D. Moneta, E. Navarro, F. G. L. Pilo, and F. Hassan, "Development and operation of active distribution networks," Tech. Rep. CIGRE WG SC6.11, 2011.
- [2] A. Hooshyar and R. Iravani, "Microgrid protection," *Proc. IEEE*, vol. 105, no. 7, pp. 1332–1353, Jul. 2017.
- [3] J. I. D. Cisneros-Saldana, S. Samal, M. M. Begovic, and S. R. Samantaray, "On protection schemes for AC microgrids: Challenges and opportunities," *IEEE Trans. Ind. Appl.*, vol. 60, no. 3, pp. 4843–4854, May 2024.
- [4] M. W. Altaf, M. T. Arif, S. N. Islam, and Md. E. Haque, "Microgrid protection challenges and mitigation approaches—A comprehensive review," *IEEE Access*, vol. 10, pp. 38895–38922, 2022.
- [5] S. C. Vegunta, M. J. Higginson, Y. E. Kenarangui, G. T. Li, D. W. Zabel, M. Tasdighi, and A. Shadman, "AC microgrid protection system design challenges—A practical experience," *Energies*, vol. 14, no. 7, p. 2016, Apr. 2021.
- [6] B. Patnaik, M. Mishra, R. C. Bansal, and R. K. Jena, "AC microgrid protection—A review: Current and future prospective," *Appl. Energy*, vol. 271, Aug. 2020, Art. no. 115210.
- [7] R. Jain, Y. N. Velaga, K. Prabakar, M. Baggu, and K. Schneider, "Modern trends in power system protection for distribution grid with high DER penetration," *e-Prime-Adv. Electr. Eng., Electron. Energy*, vol. 2, Jan. 2022, Art. no. 100080.
- [8] M. Meskin, A. Domijan, and I. Grinberg, "Impact of distributed generation on the protection systems of distribution networks: Analysis and remedies—Review paper," *IET Gener., Transmiss. Distrib.*, vol. 14, no. 24, pp. 5944–5960, Dec. 2020.
- [9] Y. Yin, Y. Fu, Z. Zhang, and A. Zamani, "Protection of microgrid interconnection lines using distance relay with residual voltage compensations," *IEEE Trans. Power Del.*, vol. 37, no. 1, pp. 486–495, Feb. 2022.
- [10] A. Sinclair, D. Finney, D. Martin, and P. Sharma, "Distance protection in distribution systems: How it assists with integrating distributed resources," *IEEE Trans. Ind. Appl.*, vol. 50, no. 3, pp. 2186–2196, May 2014.
- [11] E. Casagrande, W. L. Woon, H. H. Zeineldin, and D. Svetinovic, "A differential sequence component protection scheme for microgrids with inverter-based distributed generators," *IEEE Trans. Smart Grid*, vol. 5, no. 1, pp. 29–37, Jan. 2014.
- [12] H. Gao, J. Li, and B. Xu, "Principle and implementation of current differential protection in distribution networks with high penetration of DGs," *IEEE Trans. Power Del.*, vol. 32, no. 1, pp. 565–574, Feb. 2017.
- [13] M. J. Reno, S. Brahma, A. Bidram, and M. E. Ropp, "Influence of inverter-based resources on microgrid protection: Part 1: Microgrids in radial distribution systems," *IEEE Power Energy Mag.*, vol. 19, no. 3, pp. 36–46, May 2021.
- [14] H. J. Laaksonen, "Protection principles for future microgrids," *IEEE Trans. Power Electron.*, vol. 25, no. 12, pp. 2910–2918, Dec. 2010.
- [15] A. Hooshyar and R. Iravani, "A new directional element for microgrid protection," *IEEE Trans. Smart Grid*, vol. 9, no. 6, pp. 6862–6876, Nov. 2018.
- [16] A. Haddadi, E. Farantatos, I. Kocar, and U. Karaagac, "Impact of inverter based resources on system protection," *Energies*, vol. 14, no. 4, p. 1050, Feb. 2021.
- [17] A. Haddadi, M. Zhao, I. Kocar, U. Karaagac, K. W. Chan, and E. Farantatos, "Impact of inverter-based resources on negative sequence quantities-based protection elements," *IEEE Trans. Power Del.*, vol. 36, no. 1, pp. 289–298, Feb. 2021.
- [18] K. Opoku, S. Pokharel, and A. Dimitrovski, "A negative sequence element for fault detection in inverter-interfaced microgrids," in *Proc. IEEE Power Energy Soc. Gen. Meeting (PESGM)*, Jul. 2022, pp. 1–5.
- [19] S. Biswal and S. R. Samantaray, "Relay coordination scheme in microgrids based on non-standard tripping curve of DOCRs with multiple setting groups," *IEEE Trans. Ind. Informat.*, vol. 20, no. 4, pp. 5737–5749, Apr. 2024.

- [20] S. Biswal and S. R. Samantaray, "An effective protection coordination scheme for networked microgrids based on nonstandard tripping characteristics of DOCRs," *IEEE Syst. J.*, vol. 17, no. 4, pp. 6588–6599, Dec. 2023.
- [21] E. Dehghanpour, H. Kazemi Karegar, R. Kheirollahi, and T. Soleymani, "Optimal coordination of directional overcurrent relays in microgrids by using cuckoo-linear optimization algorithm and fault current limiter," *IEEE Trans. Smart Grid*, vol. 9, no. 2, pp. 1365–1375, Mar. 2018.
- [22] A. J. Urdaneta, L. G. Pérez, J. F. Gómez, B. Feijoo, and M. González, "Presolve analysis and interior point solutions of the linear programming coordination problem of directional overcurrent relays," *Int. J. Electr. Power Energy Syst.*, vol. 23, no. 8, pp. 819–825, Nov. 2001.
- [23] K. A. Saleh and A. Mehrizi-Sani, "Harmonic directional overcurrent relay for islanded microgrids with inverter-based DGs," *IEEE Syst. J.*, vol. 15, no. 2, pp. 2720–2731, Jun. 2021.
- [24] K. Saleh, M. A. Allam, and A. Mehrizi-Sani, "Protection of inverter-based islanded microgrids via synthetic harmonic current pattern injection," *IEEE Trans. Power Del.*, vol. 36, no. 4, pp. 2434–2445, Aug. 2021.
- [25] T. E. Sati, M. A. Azzouz, and M. F. Shaaban, "Harmonic dual-setting directional overcurrent protection for inverter-based islanded microgrids," *IEEE Access*, vol. 11, pp. 34630–34642, 2023.
- [26] P. Rajakrom, C. Booth, and Q. Hong, "Analysis and simulation of current-only directional protection incorporating simple communications," in *Proc. 57th Int. Universities Power Eng. Conf. (UPEC)*, Aug. 2022, pp. 1–6.
- [27] B. Wang and L. Jing, "A protection method for inverter-based microgrid using current-only polarity comparison," *J. Modern Power Syst. Clean Energy*, vol. 8, no. 3, pp. 446–453, May 2020.
- [28] I. Sadeghkhani, M. E. Hamedani Golshan, A. Mehrizi-Sani, J. M. Guerrero, and A. Ketabi, "Transient monitoring function-based fault detection for inverter-interfaced microgrids," *IEEE Trans. Smart Grid*, vol. 9, no. 3, pp. 2097–2107, May 2018.
- [29] D. Liu, A. Dysko, Q. Hong, D. Tzelepis, and C. D. Booth, "Transient wavelet energy-based protection scheme for inverter-dominated microgrid," *IEEE Trans. Smart Grid*, vol. 13, no. 4, pp. 2533–2546, Jul. 2022.
- [30] S. Sanati, A. Mosayebi, and I. Kamwa, "Advanced rapid directional overcurrent protection for DC microgrids using K-means clustering," *IEEE Trans. Power Del.*, vol. 39, no. 2, pp. 1088–1099, Apr. 2024.
- [31] M. Bakkar, S. Bogarra, F. Córcoles, A. Abouhassan, S. Wang, and J. Iglesias, "Artificial intelligence-based protection for smart grids," *Energies*, vol. 15, no. 13, p. 4933, Jul. 2022.
- [32] D. Li, Y. Ma, C. Zhang, H. Yin, Y. Su, L. Zhu, F. Wang, and L. M. Tolbert, "A converter-based battery energy storage system emulator for the controller testing of a microgrid with dynamic boundaries and multiple source locations," in *Proc. IEEE Energy Convers. Congr. Expo. (ECCE)*, Oct. 2022, pp. 1–8.
- [33] D. Li, Y. Su, F. Wang, M. Olama, M. Ferrari, and B. Ollis, "Modeling of a networked community microgrid with high solar penetration considering control dynamics of inverter-based resources," in *Proc. IEEE PES Innov. Smart Grid Technol. Latin Amer. (ISGT-LA)*, Nov. 2023, pp. 120–124.
- [34] A. Banaieiqadam, A. Hooshyar, and M. A. Azzouz, "A comprehensive dual current control scheme for inverter-based resources to enable correct operation of protective relays," *IEEE Trans. Power Del.*, vol. 36, no. 5, pp. 2715–2729, Oct. 2021.
- [35] A. Banaieiqadam, A. Hooshyar, and M. A. Azzouz, "A control-based solution for distance protection of lines connected to converter-interfaced sources during asymmetrical faults," *IEEE Trans. Power Del.*, vol. 35, no. 3, pp. 1455–1466, Jun. 2020.
- [36] M. Ferrari, L. M. Tolbert, and E. C. Piescirovsky, "Grid forming inverter with increased short-circuit contribution to address inverter-based microgrid protection challenges," *IEEE Open J. Ind. Electron. Soc.*, vol. 5, pp. 481–500, 2024.
- [37] Verband der Elektrotechnik Elektronik Informationstechnik VDE, *EV, Technical Requirements for The Connection and Operation of Customer Installations to the High Voltage Network (Tar High Voltage)*, Tech. Requirements Connection Operation Customer Installations High Voltage Netw. (TAR High voltage), VDE Verlag GMBH, Berlin, Germany, 2018.
- [38] J. L. Blackburn and T. J. Domin, *Protective Relaying: Principles and Applications*, Boca Raton, FL, USA: CRC Press, 2006.
- [39] A. R. V. C. Warrington, *Protective Relays: Their Theory and Practice*, vol. 1. London, U.K.: Chapman, 1962.
- [40] J. Roberts and A. Guzman, "Directional element design and evaluation," in *Proc. 21st Annu. Western Protective Relay Conf.*, Spokane, WA, USA, 1994.
- [41] E. O. Schweitzer III and J. Roberts, "Distance relay element design," in *Proc. 46th Annu. Conf. Protective Relay Engineers*, College Station, TX, USA, 1993, pp. 3–25.
- [42] B. Fleming, "Negative-sequence impedance directional element," in *Proc. 10th Annu. ProTest User Group Meeting*, 1998, pp. 1–11.
- [43] (1998). *SEL-321 Instruction Manual*. Schweitzer Engineering Laboratories, Pullman, WA, USA. [Online]. Available: <http://www.selinc.com>
- [44] B. Kasztenny, "Line distance protection near unconventional energy sources," in *Proc. 16th Int. Conf. Develop. Power Syst. Protection (DPSP)*, Mar. 2022, pp. 224–229.
- [45] B. Chen, A. Shrestha, F. A. Ituzaro, and N. Fischer, "Addressing protection challenges associated with type 3 and type 4 wind turbine generators," in *Proc. 68th Annu. Conf. Protective Relay Engineers*, Mar. 2015, pp. 335–344.
- [46] *IEEE Standard for Interconnection and Interoperability of Inverter-Based Resources (IBRS) Interconnecting With Associated Transmission Electric Power Systems*, IEEE Standard 2800-2022, 2022, pp. 1–180.
- [47] K. Opoku, A. Dimitrovski, M. Ferrari, and D. Li, "An admittance-based protection scheme for microgrids," in *Proc. IEEE PES Innov. Smart Grid Technol. Latin Amer. (ISGT-LA)*, Nov. 2023, pp. 265–269.
- [48] (2023). *SEL-411L Advanced Line Differential Protection, Automation, and Control System Instruction Manual*. Schweitzer Engineering Laboratories, Pullman, WA, USA. [Online]. Available: <http://www.selinc.com>



**KWASI OPOKU** (Graduate Student Member, IEEE) received the B.Sc. degree in electrical/electronic engineering from the Kwame Nkrumah University of Science and Technology, in 2011, and the M.S. degree in electrical engineering from the University of Central Florida, in 2023, where he is currently pursuing the Ph.D. degree in electrical engineering. He was a Protection Application Engineer with Schweitzer Engineering Laboratories Inc., and then a Service Application Engineer with GE Grid Solutions, Ghana, from 2014 to 2019. His research interests mainly include fault detection methods in inverter-interfaced microgrids and power system modeling and analysis.



**ALEKSANDAR DIMITROVSKI** (Senior Member, IEEE) received the B.Sc. degree in electrical engineering with an emphasis in power, the M.Sc. degree in applied computer sciences, and the Ph.D. degree in electrical engineering, with an emphasis in power from Europe. He is currently an Associate Professor with the Department of Electrical and Computer Engineering, University of Central Florida (UCF). Before joining UCF, he was with research institutions, the power industry, and academia, in the USA and Europe. His research interests include modeling, analysis, protection, and control of uncertain power systems.



**MAXIMILIANO FERRARI** (Senior Member, IEEE) received the B.E. degree in electronic engineering from Pontificia Universidad Javeriana, Bogotá, Colombia, in 2009, the master's degree in electronic systems engineering from the Polytechnic University of Valencia, Valencia, Spain, in 2012, the B.S. degree in physics from Allegheny College, Meadville, PA, USA, in 2015, and the Ph.D. degree in energy science and engineering from the University of Tennessee, Knoxville, TN, USA, in 2023. He was with Ford Motor Company, Intel Corporation, and Hewlett Packard. He is currently a Research and Development Associate Staff Member with the Grid Systems Architecture Group, Oak Ridge National Laboratory, Oak Ridge, TN.

...